THE MASTERMIND COACHING CLASSES
CBSE / RBSE • NURSERY – CLASS XII • JAIPUR
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ANSWER KEY & MARKING SCHEME SET 1

Chapter 3: **The World of Metals and Non-metals** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** magnesium	Basic oxide, so metal.
2	**(d)** magnesium	Magnesium is highly reactive.
3	**(b)** acids react with the metal	Acid reacts.
4	**(d)** sulphur	Non-metal.
5	**(a)** Fe	Iron.
6	**(c)** copper	Good conductor.
7	**(a)** zinc	Zinc.
8	**(a)** blue to green	$CuSO_4$ is blue.
9	**(d)** Q	Reacts fastest.
10	**(d)** less reactive than copper	Copper displaces it.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$Zn + H_2SO_4 \longrightarrow ZnSO_4 + H_2$ (1); Zinc sulphate and hydrogen (1).	1+1
12	They dissolve in rain to form acids (1); Such as SO_2 giving sulphurous acid (1).	1+1
13	They conduct heat away from the hand (1); Heat flows quickly (1).	1+1
14	Gold is very unreactive (1); Silver tarnishes with sulphur (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Metal (1); Magnesium (1); Basic oxide (1).	1+1+1
16	$MgO + H_2O \longrightarrow Mg(OH)_2$ (1); $SO_2 + H_2O \longrightarrow H_2SO_3$ (1); Basic and acidic (1).	1+1+1
17	They are brittle or soft (1); Hammers need hard metal (1); Iron or steel used (1).	1+1+1
	OR: Colour (1); Density (copper is heavier) (1); Conductivity (1).	1+1+1
18	Soft (1); Reactive with air (1); Surface tarnishes (1).	1+1+1
	OR: Sodium soft, copper hard (1); Sodium very reactive (1); Copper conducts (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Colour and lustre (1); Conduction test (1); Magnet for iron (1); Burning gives acidic or basic oxide (1); Reaction with acid (1).	1+1+1+1+1
	OR: Burn Mg ribbon (1); Dissolve ash in water (1); Burn sulphur and dissolve gas (1); Test both with litmus (1); Compare (1).	1+1+1+1+1
20	$2\,Mg + O_2 \longrightarrow 2\,MgO$ (1); $Mg + 2\,HCl \longrightarrow MgCl_2 + H_2$ (1); $Mg + CuSO_4 \longrightarrow MgSO_4 + Cu$ (1); Mg is reactive (1); Products observed (1).	1+1+1+1+1
	OR: Copper (1); Bronze alloy (1); Iron (1); Steel (1); Modern alloys (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 3: **The World of Metals and Non-metals** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** increases	Copper heavier.
2	**(b)** less reactive	They do not displace hydrogen.
3	**(a)** carbon	CO2.
4	**(d)** copper	Brown.
5	**(d)** copper sulphate	Copper ions replaced.
6	**(a)** both are forms of carbon	Allotropes.
7	**(b)** $ZnSO_4$	Iron is less reactive than zinc.
8	**(a)** making electric wires	Wires are of metals.
9	**(c)** it burns with a pop	Standard test.
10	**(b)** salt	Magnesium chloride.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Tungsten conducts and has high melting point (1); Sulphur is an insulator and melts easily (1).	1+1
12	Copper is less reactive than hydrogen (1); No displacement (1).	1+1
13	Iron is strong and stable (1); Sodium is too reactive and soft (1).	1+1
14	Good conductors (1); Carry charge to ground (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	It may be an unreactive metal (1); Protective oxide coating (1); Acid may be too dilute (1).	1+1+1
16	Oxide layer present (1); Prevents burning (1); Sand paper removes it (1).	1+1+1
17	Iron displaces copper (1); Iron sulphate is green (1); Copper deposits (1).	1+1+1
	OR: Skeleton: $Al + O_2 \longrightarrow Al_2O_3$ (1); Balance atoms (1); $4\,Al + 3\,O_2 \longrightarrow 2\,Al_2O_3$ (1).	1+1+1
18	Iron rusts in water and air (1); Coating keeps water and air away (1); Tank lasts longer (1).	1+1+1
	OR: Zinc, iron, magnesium (1); Sodium reacts violently (1); Safety (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Zn displaces Cu (1); Fe displaces Cu (1); Cu does not displace Fe (1); Mg displaces Zn (1); Mg > Zn > Fe > Cu (1).	1+1+1+1+1
	OR: Sodium soft metal (1); Diamond hard non-metal (1); Mercury liquid (1); Sulphur brittle (1); Statement false (1).	1+1+1+1+1
20	Oxygen for respiration (1); Carbon in organic compounds (1); Nitrogen in fertilisers (1); Sulphur in medicines (1); Chlorine disinfects (1).	1+1+1+1+1
	OR: Predict properties (1); Select materials (1); Study reactions (1); Organise knowledge (1); Exceptions noted (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 3

Chapter 3: **The World of Metals and Non-metals** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** it does not support burning	It is used in fire extinguishers.
2	**(c)** metal	Sonorous.
3	**(c)** calcium oxide	$CaO + H_2O \longrightarrow Ca(OH)_2$.
4	**(c)** salt and water	Neutralisation.
5	**(a)** graphite	An exceptional non-metal.
6	**(c)** sulphur dioxide	Non-metal oxide.
7	**(a)** lead	Poor among metals.
8	**(b)** different reactions with air	Different products.
9	**(a)** hydrated iron(III) oxide	Rust.
10	**(a)** protective layer and dry climate	Special composition.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Forms copper oxide (1); Black coating (1).	1+1
12	Add the oxide solution to turmeric (1); It turns red (1).	1+1
13	Acid reacts with it (1); May form harmful compounds (1).	1+1
14	Liquid metal (1); Expands evenly (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Harder (1); Resist corrosion (1); Cost control (1).	1+1+1
16	Metals expand on heating (1); Wire becomes longer (1); Longer wire hangs lower (1).	1+1+1
17	It does not support burning (1); Heavier than air (1); Covers the fire (1).	1+1+1
	OR: Move magnet over the mixture (1); Iron filings stick (1); Sulphur is left (1).	1+1+1
18	Zinc is more reactive (1); It corrodes first (1); Protects iron (1).	1+1+1
	OR: Mixtures of metals (1); Brass: copper and zinc (1); Steel: iron and carbon (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Rusting of bridges (1); Tarnishing of silver (1); Loss of money (1); Paint and galvanising (1); Regular maintenance (1).	1+1+1+1+1
	OR: Glows and a black solid forms (1); Not attracted by a magnet (1); Does not dissolve in solvent like sulphur (1); Properties differ (1); Iron sulphide is formed (1).	1+1+1+1+1
20	Tests show metals light the bulb (1); Wax melts along a metal rod (1); Non-metals do not (1); Graphite is an exception (1); Use decides (1).	1+1+1+1+1
	OR: Saves ores (1); Saves energy (1); Reduces waste (1); Reduces pollution (1); Creates jobs (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.